

COHERENT CONDITIONAL PREFERENCES AND TWO-ENVELOPE PARADOXES

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Subjective expected utility theory is predicated on:

- ◇ a state-space that represents resolutions of uncertainty
- ◇ 'acts' whose payoff depends on the state

By contemplating such acts, the DM reasons contingently about the the likelihood of states

Savage writes (1954, p. 16),

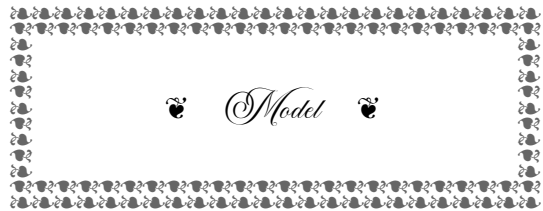


ARRIED to its logical extreme, [contemplating acts] demands that one envisage every conceivable policy for the government of his whole life in its most minute details, in the light of the vast number of unknown states of the world, and decide here and now on one policy. This is utterly ridiculous, not—as some might think—because there might later be cause for regret, if things did not turn out as had been anticipated, but because the task implied in making such a decision is not even remotely resembled by human possibility.

- ◇ Savage argues SEU should apply only in *small worlds*
 - ◇ Where the state-space can be fully understood
- ◇ Small worlds might intersect
- ◇ Preferences should be consistent on the intersection

We propose a theory of small worlds

- ◇ Dynamic consistency turns out to be insufficient to shield from money pump schemes
 - ◇ These take the form of ‘two-envelope paradoxes’
- ◇ Coherency requires *global* consistency
 - ◇ This undermines the appeal of the ‘small-worlds’ approach



- ◇ Let $(\mathcal{S}, \mathfrak{A})$ be a measurable space (the universe)
- ◇ $\mathfrak{B} \subseteq \mathfrak{A} \setminus \{\emptyset\}$ are *small worlds*
 - ◇ closed under intersections and finite unions
 - ◇ $\bigcup \mathfrak{B} = \mathcal{S}$
 - ◇ $\mathcal{S} \notin \mathfrak{B}$

Examples:

- ◇ $\mathcal{S}$ is infinite, $\mathfrak{B}$ is finite subsets.
- ◇ $(\mathcal{S}, \mathfrak{A}, \mu)$ is an infinite measure space, and $\mathfrak{B} \subset \mathfrak{A}$ is finite μ -measure subsets
- ◇ $\mathcal{S}$ is an unbounded metric space, $\mathfrak{B}$ is bounded open subsets
- ◇ $\mathcal{S}$ is a lattice with no maximal element, $\mathfrak{B}$ is subsets that are bounded above.

- ◇ $(\mathcal{X}, \mathfrak{C})$ is a measurable space of outcomes
- ◇ $\mathcal{F}$ are measurable functions (“acts”) from $\mathcal{S}$ to $\mathcal{X}$.
- ◇ Our primitive is a **conditional preference structure**:
 - ◇ a collection $\{\succsim_B\}_{B \in \mathfrak{B}}$ of weak orders on $\mathcal{F}$,

A0: Small World Expected Utility

For all $\mathcal{B} \in \mathfrak{B}$, $\succsim_{\mathcal{B}}$ is an expected utility preference that regards $\mathcal{S} \setminus \mathcal{B}$ as null.
That is: for all $f, g \in \mathcal{F}$:

$$f|_{\mathcal{B}} = g|_{\mathcal{B}} \implies f \approx_{\mathcal{B}} g$$

A1: Ex-Post Consistency

For all $\mathcal{B}, \mathcal{C} \in \mathfrak{B}$, $\succsim_{\mathcal{B}}$ and $\succsim_{\mathcal{C}}$ rank constant acts in same way

A2: Dynamic Consistency

For all $\mathcal{B}, \mathcal{C} \in \mathfrak{B}$ and $f, g \in \mathcal{F}$: if $\mathcal{B} \subseteq \mathcal{C}$ and $f|_{(\mathcal{C} \setminus \mathcal{B})} = g|_{(\mathcal{C} \setminus \mathcal{B})}$ then

$$f \succcurlyeq_{\mathcal{B}} g \quad \Longleftrightarrow \quad f \succcurlyeq_{\mathcal{C}} g$$

Theorem

The following are equivalent:

- (1) $\{\succsim_{\mathcal{B}}\}_{\mathcal{B} \in \mathfrak{B}}$ satisfies A0–A2
- (2) There exists a utility $u : \mathcal{X} \rightarrow \mathbb{R}$ and a measure ρ over $(\mathcal{S}, \mathfrak{A})$ such that for all $f, g \in \mathcal{F}$, $f \succsim_{\mathcal{B}} g$ if and only if

$$\int_{\mathcal{B}} u \circ f \, d\rho(\cdot \mid \mathcal{B}) \geq \int_{\mathcal{B}} u \circ g \, d\rho(\cdot \mid \mathcal{B})$$

- ◇ $\rho(\mathcal{B})$ is finite for all $\mathcal{B} \in \mathfrak{B}$
- ◇ $\rho(\cdot \mid \mathcal{B})$ is the conditional probability: $\rho(\mathcal{A} \mid \mathcal{B}) = \frac{\rho(\mathcal{A} \cap \mathcal{B})}{\rho(\mathcal{B})}$

This seems ultra rational:

- ◇ Dynamic consistency is a local notion
- ◇ Allows us to glue together different small worlds
- ◇ Expected utility via conditioning on some global perspective

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- ◇ Dynamic consistency is a local notion
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So what can go wrong?



The classic two envelope paradox:



HERE are two envelopes: 'Lucky' and 'Unlucky'. Alice puts $x \geq 1$ dollars into the Unlucky envelope and puts exactly twice that amount in the Lucky envelope. She then gives one of the envelopes to Bob; Bob doesn't know which envelope he received. If Bob were to open the his current envelope and see $x \geq 2$ inside, he would believe there is an equal chance that he holds the Lucky or the Unlucky envelope. Before he opens the envelope, Bob can choose to KEEP it or SWITCH it with the other one.

We can think of this via 'small worlds'

- ◇ It may not be possible to consider the probability of all possible values ex-ante
- ◇ But, once the envelope is opened, and he sees $\$x$, it is a small world:
- ◇ There are only two possibilities:
 - (1) Unlucky envelope $\$x$, Lucky envelope $\$2x$
 - (2) Unlucky envelope $\$\frac{1}{2}x$, Lucky envelope $\$x$

- ◇ If Bob opens the envelope and sees $\$x \geq 2$, the expected value of switching is

$$\underbrace{\frac{1}{2} \left(\frac{1}{2}x \right)}_{\text{lucky}} + \underbrace{\frac{1}{2} \left(\frac{1}{2}2x \right)}_{\text{unlucky}} = \frac{5}{2}x$$

- ◇ If Bob opens the envelope and sees $\$x < 2$, definitely better to SWITCH
- ◇ Thus, independent of x , he is better off with SWITCH (assume risk neutral)
- ◇ This is a problem: charge a little bit ex-ante to SWITCH $\rightarrow$ money pump

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U	w_{1U}	w_{2U}	w_{4U}	w_{8U}	w_{16U}	$\dots$
L	w_{1L}	w_{2L}	w_{4L}	w_{8L}	w_{16L}	$\dots$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U L	w_{1U}	w_{2U}	w_{4U}	w_{8U}	w_{16U}	$\dots$
	w_{1L}	w_{2L}	w_{4L}	w_{8L}	w_{16L}	$\dots$
KEEP	1	2	4	8	16	$\dots$
	2	4	8	16	32	$\dots$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U L	w_{1U}	w_{2U}	w_{4U}	w_{8U}	w_{16U}	$\dots$
	w_{1L}	w_{2L}	w_{4L}	w_{8L}	w_{16L}	$\dots$
KEEP	1	2	4	8	16	$\dots$
	2	4	8	16	32	$\dots$
SWITCH	2	4	8	16	32	$\dots$
	1	2	4	8	16	$\dots$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U L	w_{1U}	w_{2U}	w_{4U}	w_{8U}	w_{16U}	...
	w_{1L}	w_{2L}	w_{4L}	w_{8L}	w_{16L}	...
KEEP	1	2	4	8	16	...
	2	4	8	16	32	...
SWITCH	2	4	8	16	32	...
	1	2	4	8	16	...

$A_0 = \text{'\$1 in Unlucky'} - \text{KEEP} \sim_{A_0} \text{SWITCH}$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U	$\frac{1}{2}$	w_{2U}	w_{4U}	w_{8U}	w_{16U}	...
L	$\frac{1}{2}$	w_{2L}	w_{4L}	w_{8L}	w_{16L}	...
KEEP	1	2	4	8	16	...
	2	4	8	16	32	...
SWITCH	2	4	8	16	32	...
	1	2	4	8	16	...

$A_1 = \text{'\$2 in Unlucky'} - \text{KEEP} \sim_{A_1} \text{SWITCH}$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U L	w_{1U}	$\frac{1}{2}$	w_{4U}	w_{8U}	w_{16U}	$\dots$
	w_{1L}	$\frac{1}{2}$	w_{4L}	w_{8L}	w_{16L}	$\dots$
KEEP	1	2	4	8	16	$\dots$
	2	4	8	16	32	$\dots$
SWITCH	2	4	8	16	32	$\dots$
	1	2	4	8	16	$\dots$

$A_2 = \text{'\$4 in Unlucky'} - \text{KEEP} \sim_{A_2} \text{SWITCH}$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U L	w_{1U}	w_{2U}	$\frac{1}{2}$	w_{8U}	w_{16U}	$\dots$
	w_{1L}	w_{2L}	$\frac{1}{2}$	w_{8L}	w_{16L}	$\dots$
KEEP	1	2	4	8	16	$\dots$
	2	4	8	16	32	$\dots$
SWITCH	2	4	8	16	32	$\dots$
	1	2	4	8	16	$\dots$

$A_3 = \text{'\$8 in Unlucky'} - \text{KEEP} \sim_{A_3} \text{SWITCH}$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U L	w_{1U}	w_{2U}	w_{4U}	$\frac{1}{2}$	w_{16U}	...
	w_{1L}	w_{2L}	w_{4L}	$\frac{1}{2}$	w_{16L}	...
KEEP	1	2	4	8	16	...
	2	4	8	16	32	...
SWITCH	2	4	8	16	32	...
	1	2	4	8	16	...

$$A_4 = \text{'\$16 in Unlucky'} - \text{KEEP} \sim_{A_4} \text{SWITCH}$$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U	w_{1U}	w_{2U}	w_{4U}	w_{8U}	$\frac{1}{2}$	...
	w_{1L}	w_{2L}	w_{4L}	w_{8L}	$\frac{1}{2}$	...
KEEP	1	2	4	8	16	...
	2	4	8	16	32	...
SWITCH	2	4	8	16	32	...
	1	2	4	8	16	...

Let φ be the ‘given the opposite envelope to Bob operator’

$$\varphi : \begin{cases} w_{iU} \mapsto w_{iL} \\ w_{iL} \mapsto w_{iU} \end{cases}$$

Then $\text{SWITCH} = \text{KEEP} \circ \varphi$

- ◇ φ is belief preserving for all conditional probabilities
- ◇ The events $\{A_i\}_{i \in \mathbb{N}}$ are φ -invariant
- ◇ EU demands $\text{SWITCH} \sim_{A_i} \text{KEEP}$ for all i

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U L	w_{1U}	w_{2U}	w_{4U}	w_{8U}	w_{16U}	$\dots$
	w_{1L}	w_{2L}	w_{4L}	w_{8L}	w_{16L}	$\dots$
KEEP	1	2	4	8	16	$\dots$
	2	4	8	16	32	$\dots$
SWITCH	2	4	8	16	32	$\dots$
	1	2	4	8	16	$\dots$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U L	w_{1U}	w_{2U}	w_{4U}	w_{8U}	w_{16U}	...
	w_{1L}	w_{2L}	w_{4L}	w_{8L}	w_{16L}	...
KEEP	1	2	4	8	16	...
	2	4	8	16	32	...
SWITCH	2	4	8	16	32	...
	1	2	4	8	16	...

$B_0 = \text{'Observe \$1'} - \text{SWITCH} \succ_{B_0} \text{KEEP}$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U	1	w_{2U}	w_{4U}	w_{8U}	w_{16U}	...
L	w_{1L}	w_{2L}	w_{4L}	w_{8L}	w_{16L}	...
KEEP	1	2	4	8	16	...
	2	4	8	16	32	...
SWITCH	2	4	8	16	32	...
	1	2	4	8	16	...

$B_1 = \text{'Observe \$2'} - \text{SWITCH} \succ_{B_1} \text{KEEP}$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U	w_{1U}	$\frac{1}{2}$	w_{4U}	w_{8U}	w_{16U}	...
L	$\frac{1}{2}$	w_{2L}	w_{4L}	w_{8L}	w_{16L}	...
KEEP	1	2	4	8	16	...
	2	4	8	16	32	...
SWITCH	2	4	8	16	32	...
	1	2	4	8	16	...

$B_2 = \text{'Observe \$4'} - \text{SWITCH} \succ_{B_2} \text{KEEP}$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U L	w_{1U}	w_{2U}	$\frac{1}{2}$	w_{8U}	w_{16U}	...
	w_{1L}	$\frac{1}{2}$	w_{4L}	w_{8L}	w_{16L}	...
KEEP	1	2	4	8	16	...
	2	4	8	16	32	...
SWITCH	2	4	8	16	32	...
	1	2	4	8	16	...

$B_3 = \text{'Observe \$8'} - \text{SWITCH} \succ_{B_3} \text{KEEP}$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U L	w_{1U}	w_{2U}	w_{4U}	$\frac{1}{2}$	w_{16U}	...
	w_{1L}	w_{2L}	$\frac{1}{2}$	w_{8L}	w_{16L}	...
KEEP	1	2	4	8	16	...
	2	4	8	16	32	...
SWITCH	2	4	8	16	32	...
	1	2	4	8	16	...

$B_4 = \text{'Observe \$16'} - \text{SWITCH} \succ_{B_4} \text{KEEP}$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U L	w_{1U}	w_{2U}	w_{4U}	w_{8U}	$\frac{1}{2}$	...
	w_{1L}	w_{2L}	w_{4L}	$\frac{1}{2}$	w_{16L}	...
KEEP	1	2	4	8	16	...
	2	4	8	16	32	...
SWITCH	2	4	8	16	32	...
	1	2	4	8	16	...

- ◇ There exists a preserving map, φ , and two partitions of the state-space:
 - ◇ $\{A_i\}_{i \in \mathbb{N}}$ — the amount put in the Unlucky envelope is 2^i
 - ◇ $\{B_i\}_{i \in \mathbb{N}}$ — the amount observed in the given envelope is 2^i
- ◇ The small worlds, A_i , are φ -invariant (so $\text{SWITCH} \sim_{B_i} \text{KEEP}$)
- ◇ In every B_i small world, $\text{SWITCH} \succ_{B_i} \text{KEEP}$
- ◇ Call such a state-of-affairs a **generalized two-envelope paradox**

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U	w_{1U}	w_{2U}	w_{4U}	w_{8U}	w_{16U}	$\dots$
L	w_{1L}	w_{2L}	w_{4L}	w_{8L}	w_{16L}	$\dots$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U	$\rho = \alpha$	w_{2U}	w_{4U}	w_{8U}	w_{16U}	$\dots$
L	w_{1L}	w_{2L}	w_{4L}	w_{8L}	w_{16L}	$\dots$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U	$\rho_{A_0} = \frac{1}{2}$	w_{2U}	w_{4U}	w_{8U}	w_{16U}	$\dots$
L	$\rho_{A_0} = \frac{1}{2}$	w_{2L}	w_{4L}	w_{8L}	w_{16L}	$\dots$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U	$\rho = \alpha$	w_{2U}	w_{4U}	w_{8U}	w_{16U}	$\dots$
L	$\rho = \alpha$	w_{2L}	w_{4L}	w_{8L}	w_{16L}	$\dots$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U	w_{1U}	$\rho_{B_1} = \frac{1}{2}$	w_{4U}	w_{8U}	w_{16U}	$\dots$
L	$\rho_{B_1} = \frac{1}{2}$	w_{2L}	w_{4L}	w_{8L}	w_{16L}	$\dots$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U	$\rho = \alpha$	$\rho = \alpha$	w_{4U}	w_{8U}	w_{16U}	$\dots$
L	$\rho = \alpha$	w_{2L}	w_{4L}	w_{8L}	w_{16L}	$\dots$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U	w_{1U}	w_{2U}	$\rho_{B_2} = \frac{1}{2}$	w_{8U}	w_{16U}	$\dots$
L	w_{1L}	$\rho_{B_2} = \frac{1}{2}$	w_{4L}	w_{8L}	w_{16L}	$\dots$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U	$\rho = \alpha$	$\rho = \alpha$	$\rho = \alpha$	w_{8U}	w_{16U}	$\dots$
L	$\rho = \alpha$	$\rho = \alpha$	w_{4L}	w_{8L}	w_{16L}	$\dots$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U	w_{1U}	w_{2U}	w_{4U}	$\rho_{B_3} = \frac{1}{2}$	w_{16U}	$\dots$
L	w_{1L}	w_{2L}	$\rho_{B_3} = \frac{1}{2}$	w_{8L}	w_{16L}	$\dots$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U	$\rho = \alpha$	$\rho = \alpha$	$\rho = \alpha$	$\rho = \alpha$	w_{16U}	$\dots$
L	$\rho = \alpha$	$\rho = \alpha$	$\rho = \alpha$	w_{8L}	w_{16L}	$\dots$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U	w_{1U}	w_{2U}	w_{4U}	w_{8U}	$\rho_{B_4} = \frac{1}{2}$	$\dots$
L	w_{1L}	w_{2L}	w_{4L}	$\rho_{B_4} = \frac{1}{2}$	w_{16L}	$\dots$

	$x = 1$	$x = 2$	$x = 4$	$x = 8$	$x = 16$	
U	$\rho = \alpha$	$\rho = \alpha$	$\rho = \alpha$	$\rho = \alpha$	$\rho = \alpha$	...
L	$\rho = \alpha$	$\rho = \alpha$	$\rho = \alpha$	$\rho = \alpha$	$\rho = \alpha$	...

- ◇ The two-envelope paradox leads to an infinite ρ
- ◇ Cannot be normalized to a probability measure
- ◇ Is this the 'source' of the paradox?

The variant of the two envelope paradox:



LICE puts $x \geq 1$ dollars into the Unlucky envelope and then puts exactly three times that amount in the Lucky envelope. She then gives one of the two envelopes to Bob; Bob doesn't know which envelope he received. If Bob were to open the his current envelope and see $x \geq 3$ inside, he would believe there is a $\frac{2}{3}$ chance that he holds the Lucky and a $\frac{1}{3}$ chance he holds the Unlucky envelope.

	$x = 1$	$x = 3$	$x = 9$	$x = 27$	$x = 81$	
U	$\rho = \frac{1}{4}$	$\rho = \frac{1}{8}$	$\rho = \frac{1}{16}$	$\rho = \frac{1}{32}$	$\rho = \frac{1}{64}$	$\dots$
L	$\rho = \frac{1}{4}$	$\rho = \frac{1}{8}$	$\rho = \frac{1}{16}$	$\rho = \frac{1}{32}$	$\rho = \frac{1}{64}$	$\dots$
KEEP	1	3	9	27	81	$\dots$
	3	9	27	81	243	$\dots$
SWITCH	3	9	27	81	243	$\dots$
	1	3	9	27	81	$\dots$

Theorem

Let $\{\succsim_B\}_{B \in \mathfrak{B}}$ be a cond. preference structure with a EU representation (u, ρ) .

Then $\{\succsim_B\}_{B \in \mathfrak{B}}$ admits a gen. two-envelope paradox if and only if either

(1) $\sup_{x \in \mathcal{X}} |u(x)| = \infty$; or

(2) $\rho(\mathcal{S}) = \infty$.



♥ *Thank You* ♥